Knowledge Bases

EECS 498 AASE, Lecture 17

Answers rebuilt from scratch

Monday

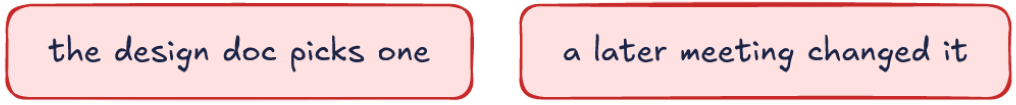


a similar question on Thursday



the same search puts the same pieces together again

two of the documents



the answer may use the older one

How does knowledge build up instead of being found again?

Today's schedule

- [x] Building a knowledge base
- [ ] Break
- [x] Finding and trusting it
- [ ] Questions

Part 1

Building a knowledge base

Who keeps the wiki

a wiki people keep



the upkeep grows faster than the value

you

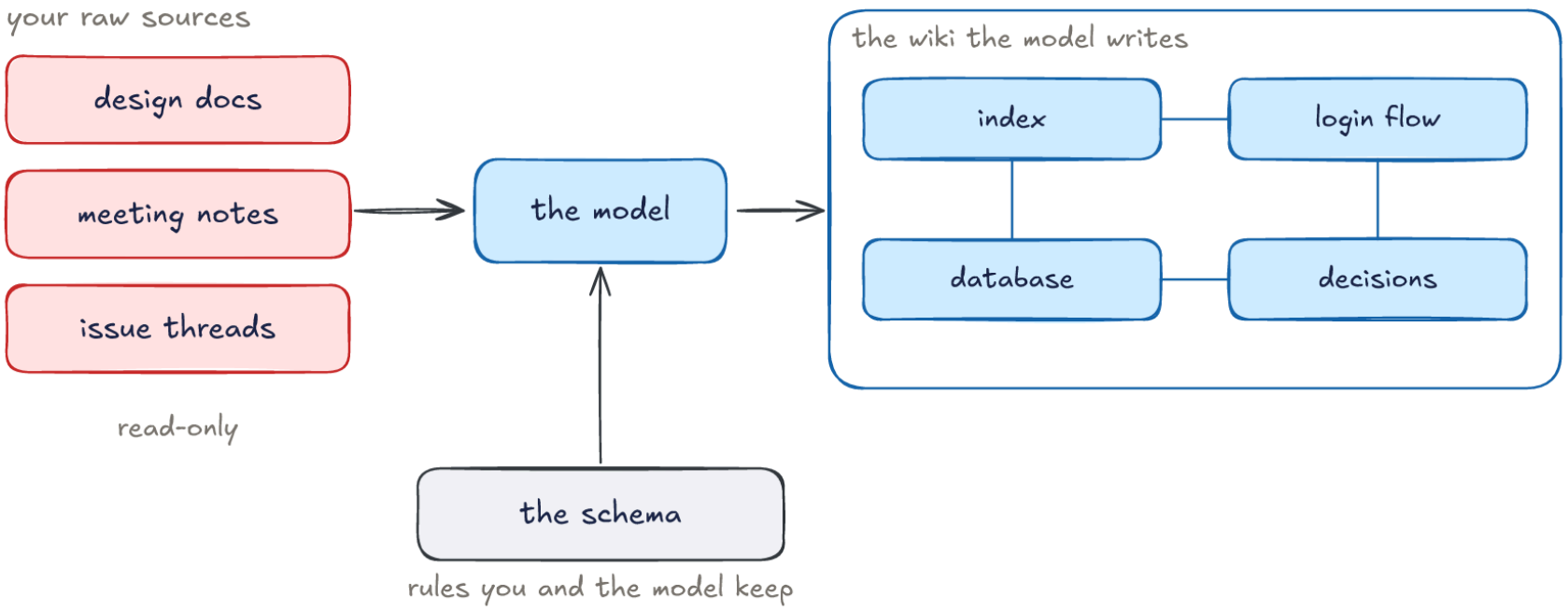
- pick the sources
- ask the questions
- decide what matters

the model

- summaries and links
- the index and the log
- keeping pages consistent

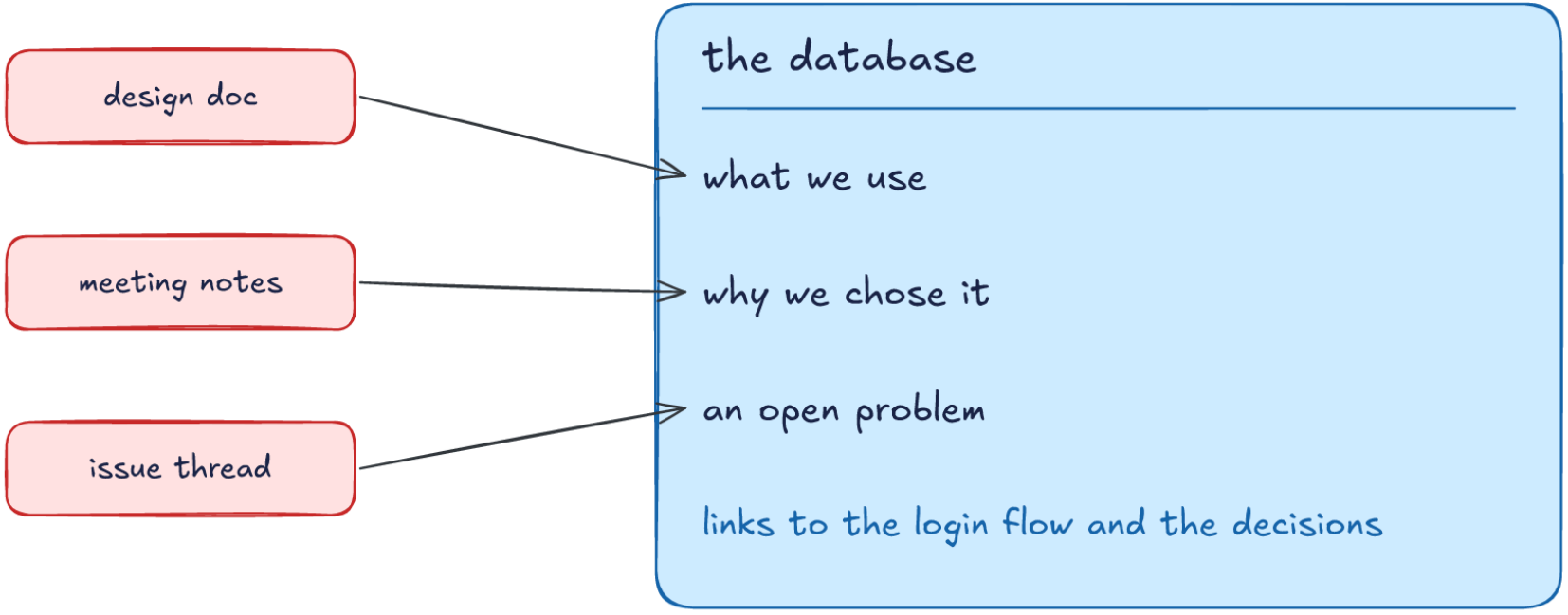
the bookkeeping now costs the model almost nothing

Three layers



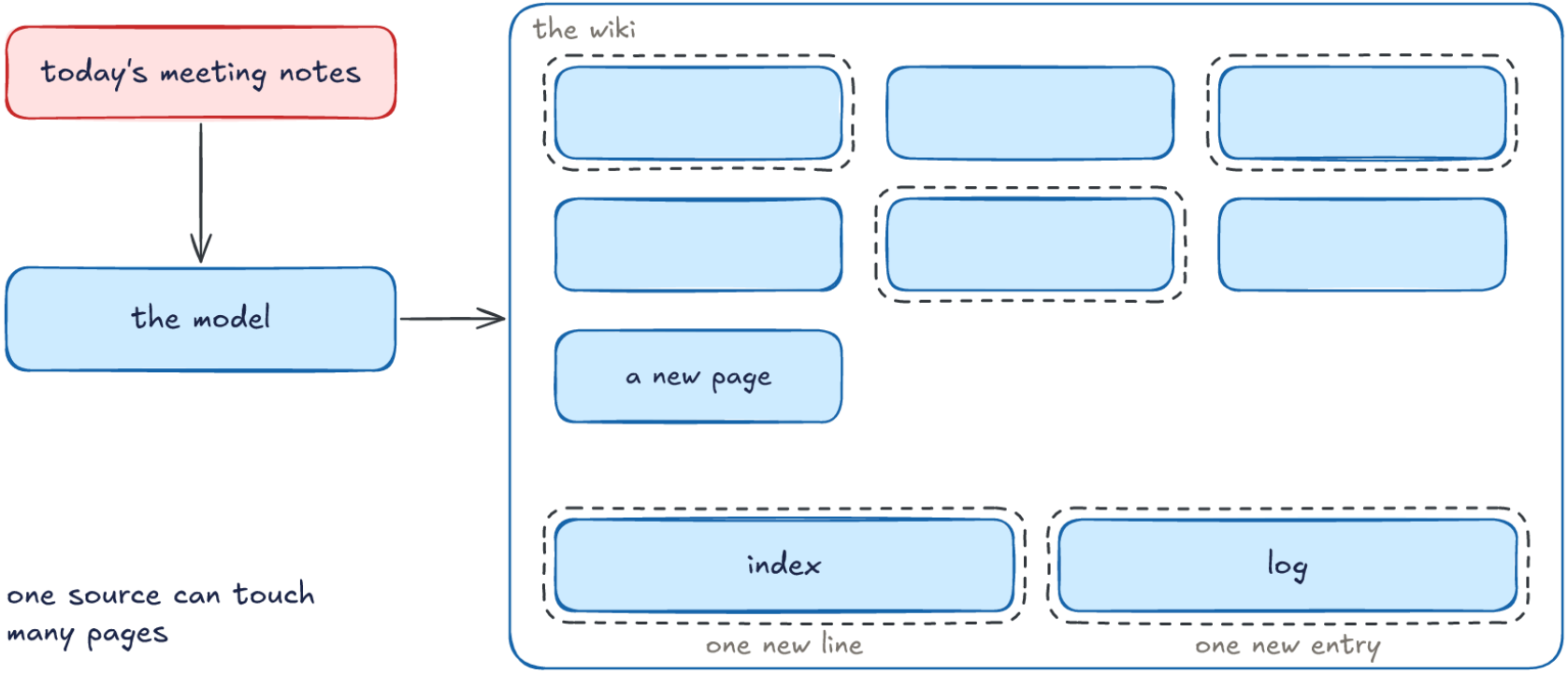
memory learns from sessions and a wiki learns from sources

One page from many sources



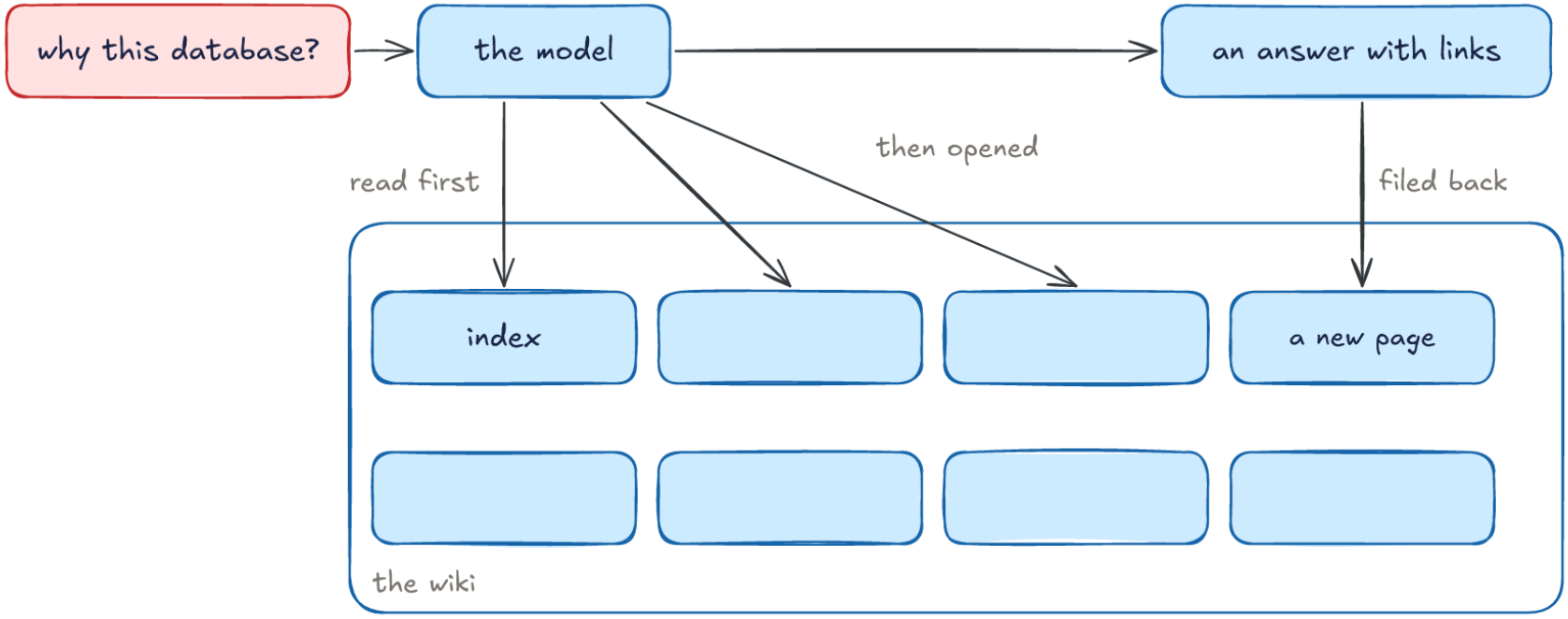
the page sums up several sources instead of copying one

Adding a source

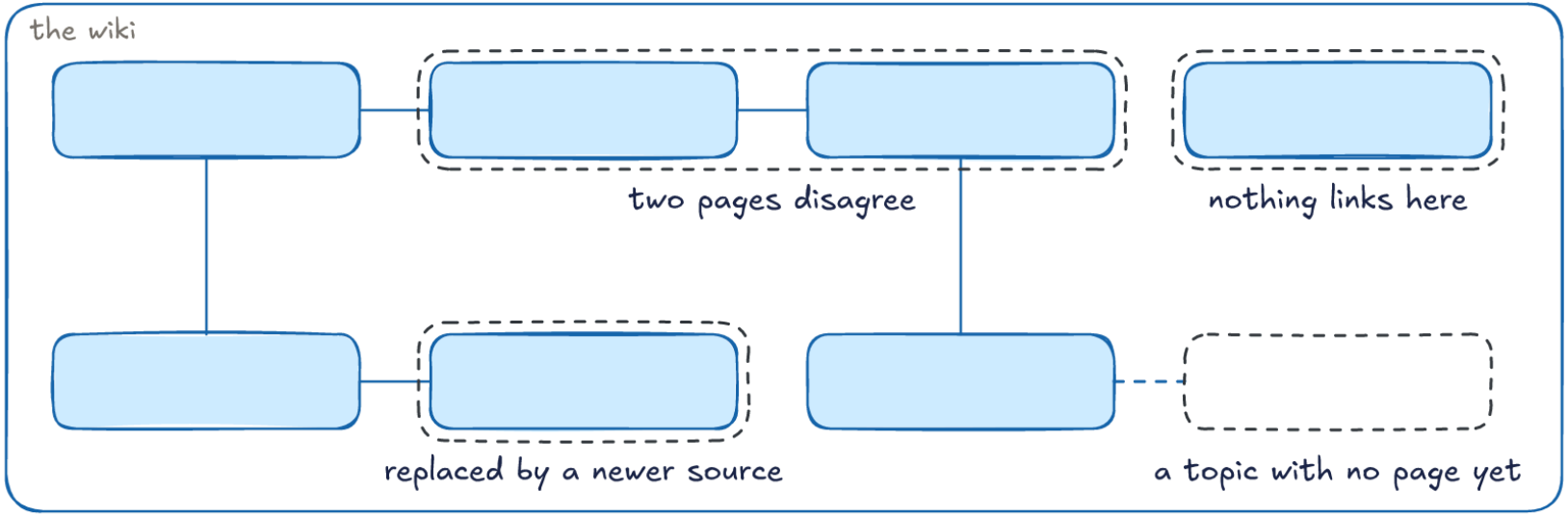


one source can touch
many pages

Asking a question



Checking the wiki



then it fixes the pages or asks you which source is right

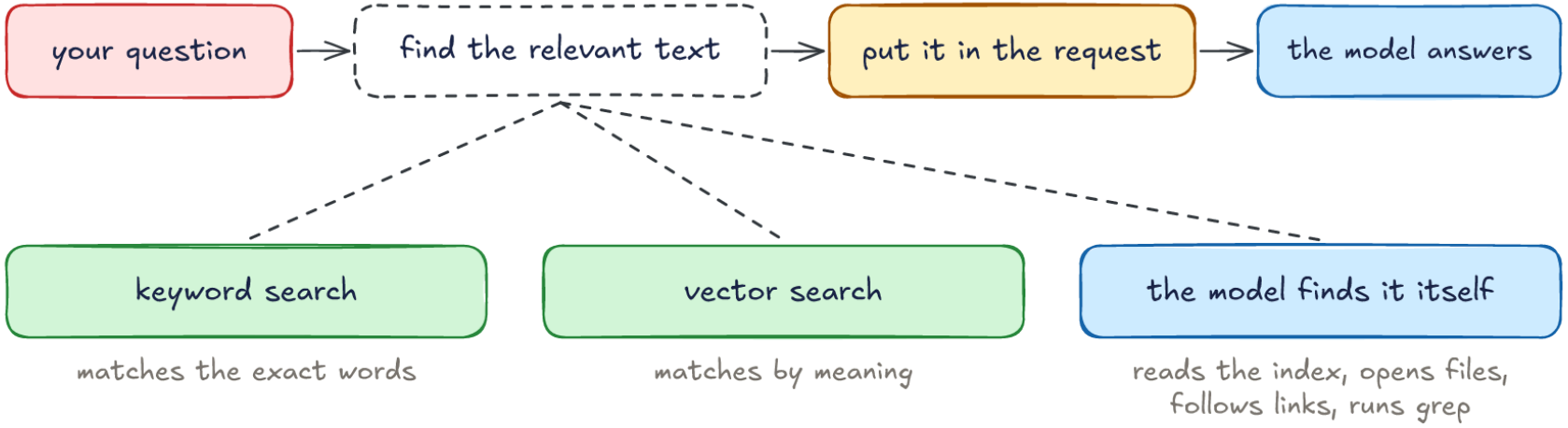
Chat break

Back in five minutes

Part 2

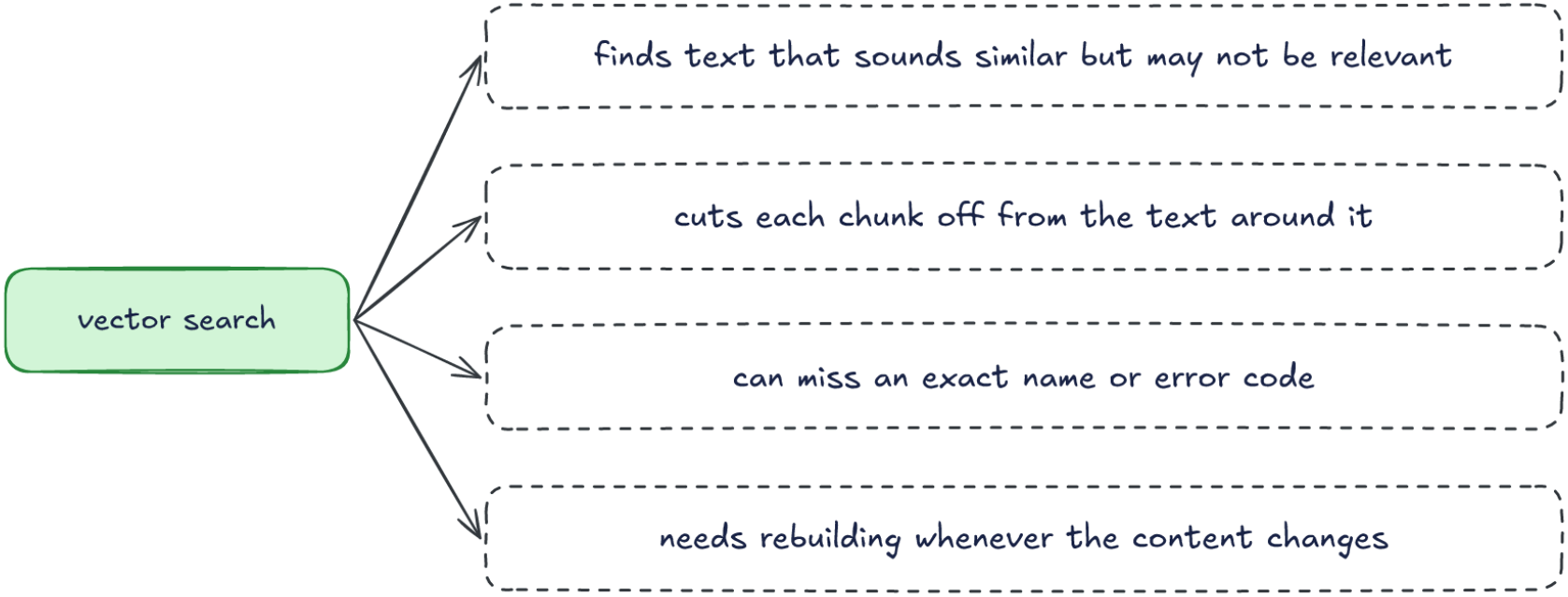
Finding and trusting it

Finding the relevant text



retrieval is the job and vector search is one tool for it

Limits of vector search



Retrieve or compile

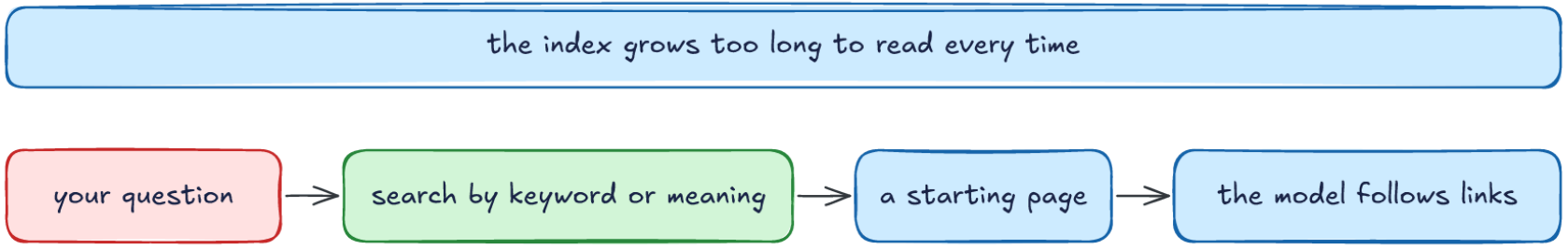
	retrieve from raw documents	compile a wiki
the work happens	at every question	when a source arrives
what is kept	nothing	pages that build up
what the model gets	pieces of the raw documents	pages written with their context
it fits	a big collection nobody will curate	knowledge you work with over time

When the index is not enough

hundreds of pages

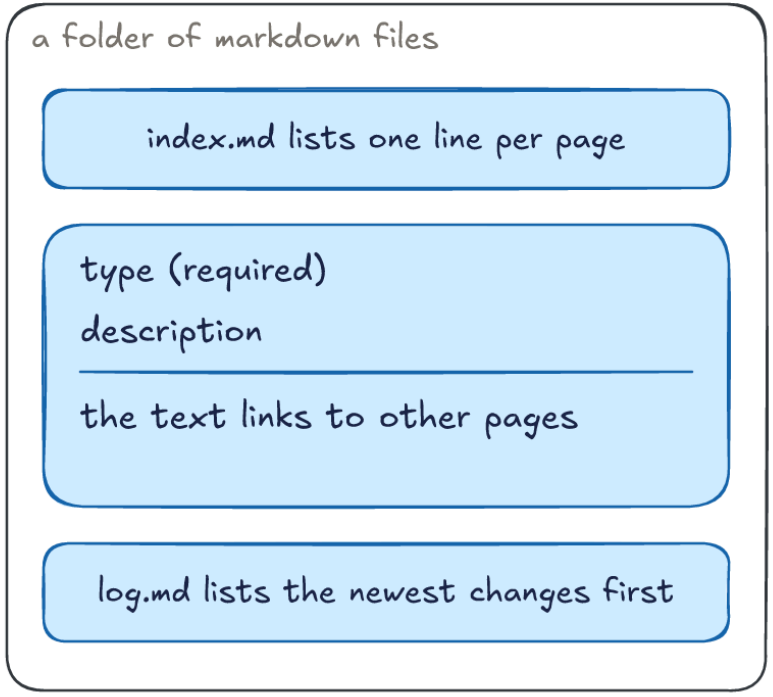


thousands of pages



search finds a starting point and links do the rest

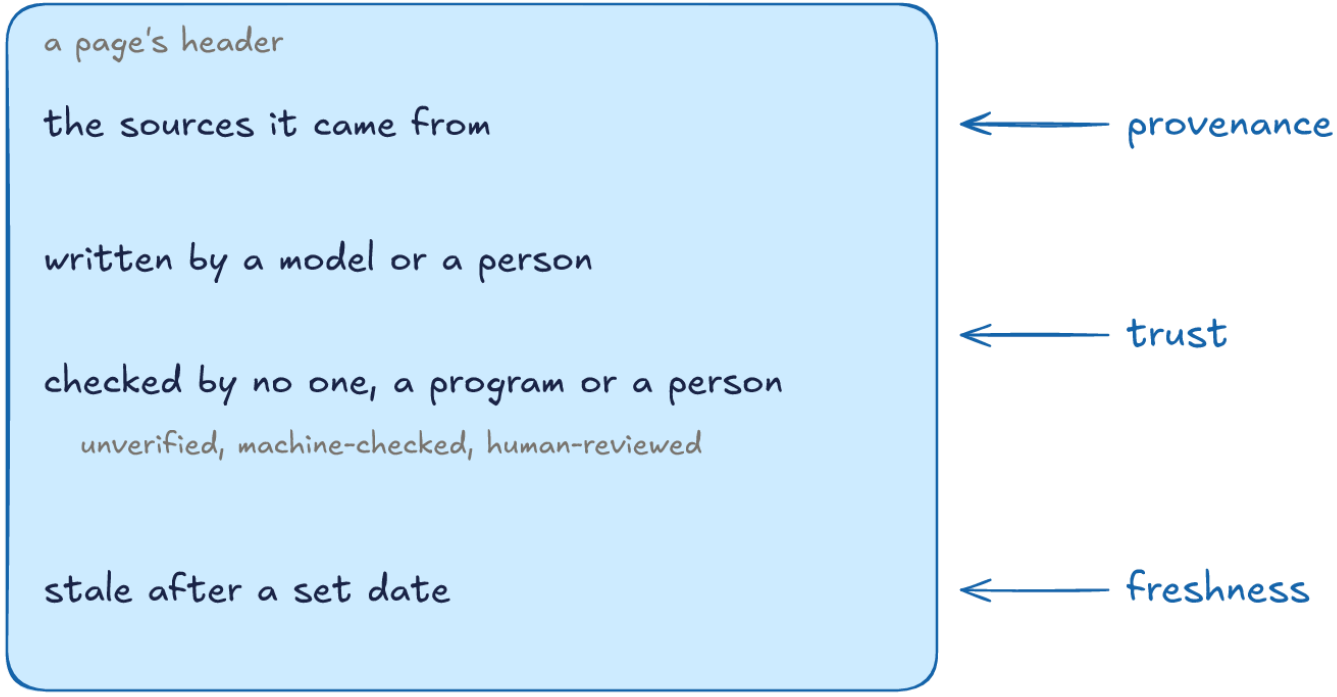
An open format



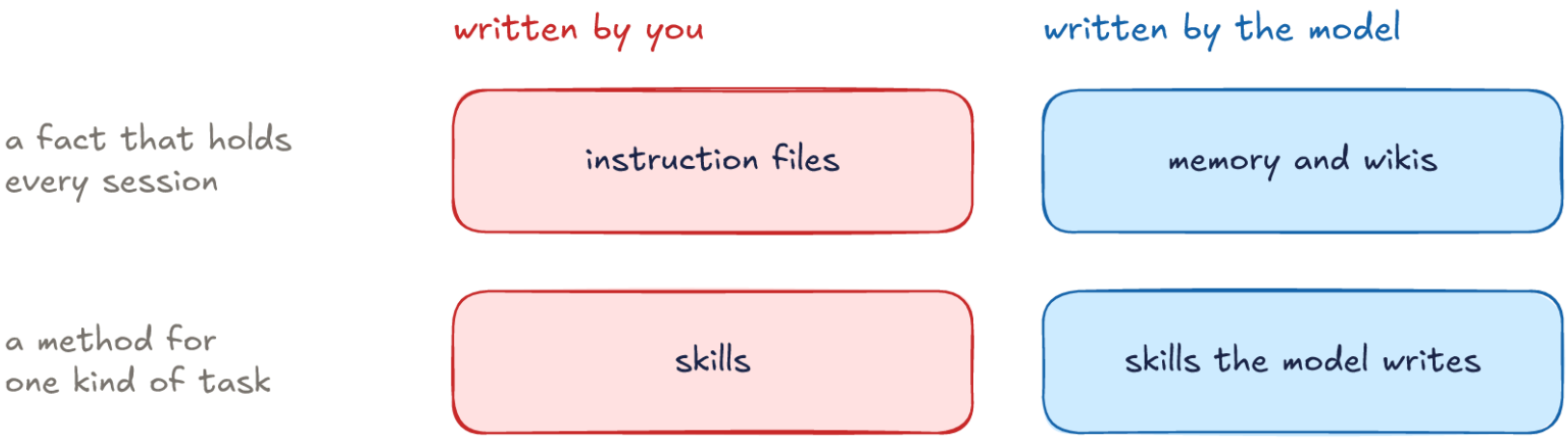
every harness could invent
its own layout

any harness that reads files
can read it

Trust and freshness



Where knowledge lives



a skill's description, the memory index and the wiki's index.md are one mechanism, so a harness needs one way to load them

What to take away

- A wiki compiles sources and keeps them current
- You pick sources and the model keeps the pages
- Retrieve when nobody will curate
- A page records its source and its checker

Sources

Karpathy's LLM Wiki
gist.github.com/karpathy/442a6bf555914893e9891c11519de94f

The Open Knowledge Format version 0.2
github.com/GoogleCloudPlatform/knowledge-catalog

Anthropic on contextual retrieval
anthropic.com/news/contextual-retrieval

The RAG paper by Lewis and others (2020)
arxiv.org/abs/2005.11401

Questions

AASE Lecture 17, Knowledge Bases